

Exploration Sprint

For curiosity-driven "what ifs" that aren't research questions yet. This is the precursor to the Research Question Sharpener — use it to find out if there's something worth sharpening.

BEFORE YOU START

The What-If

What are you curious about? State it naturally. "What would happen if..." or "I wonder whether..." is fine here.

I want to see how far the new frontier models/agent approaches can get me on a relatively hard problem.

To describe the problem: we want to find all possible absolute determinant values for a 12×12 matrix where every entry is 1 or -1. By Hadamard's inequality, the max det is $12^6 = 2,985,984$, and it must be a multiple of $2^{11} = 2048$, leaving exactly 1,458 possible positive values it could take. This problem is actually solved for all $n < 12$ and $n = 13$.

The raw search space is a massive 2^{144} . While symmetry can reduce this, the normal literature approach (which solved $n = 13$) is to search the tree of Gram Matrices ($G = AA^T$) and prune branches using the Positive Semi-Definite (PSD) constraint. But for $n = 12$, this does not work. Because 12 is even, the dot product of two rows can be 0 (perfectly orthogonal). These zeros bypass some constraints, causing the search space to go insane.

What if we just point a state-of-the-art agent loop at the problem and see if it can figure out how to navigate the space and set new bounds? $n = 14$ might be similar in difficulty as it's not a multiple of 4 and that will probably reduce some search space

Why this, why now?

What prompted this curiosity? A paper you read, an observation, a conversation, a hunch? Capturing the origin helps you notice what kinds of intuitions tend to be productive.

This line: "A related (and even more difficult) problem is the determinant spectrum problem which asks, not just for the maximal determinant, but for the complete set of values taken by the determinant function." here:

<https://web.archive.org/web/20200218010153/http://www.indiana.edu/~maxdet/fullPage.shtml#tableTo>
[p](#)

Also,

With all the recent news over the last few months of GPT-5.4/5.5 and Harmonic's Aristotle solving long-standing Erdős problems (Harmonic solving Erdős problems: <https://arxiv.org/abs/2601.07421>)

and clearing benchmarks like (Epoch AI's FrontierMath benchmark: <https://epochai.org/frontiermath>), I think these systems are now finally at a level where they are making actually interesting mathematical discoveries.

The determinant spectrum for $n=12$ $\{-1, 1\}$ matrices has been stalled for 10+ years. For context on the limits of current exhaustive searches, see the completion of the $n=13$ spectrum and lower bounds by Brent and Osborn in their paper (<https://arxiv.org/abs/1112.4160>).

I just did one small probing session with Google Gemini 3.1 Pro, and it already gave me a couple of different approaches to move forward with:

- Using a SAT+CAS (Satisfiability + Computer Algebra System) architecture.
- Injecting modulo-4 parity constraints ($G_{\{ij\}} \equiv -2(r_i + r_j) \pmod{4}$) directly into the solver to aggressively prune the 2^{144} search space natively.

I don't completely agree with it as of now, but it is atleast different from what is used uptill now.

EXPECTATIONS

Without written expectations, you can't recognize surprise — and surprise is what turns exploration into research.

What do you expect to observe?

Be specific. "I expect that scaling cellular automata 100x will produce qualitatively similar dynamics" or "I expect performance to degrade gradually, not suddenly." Vague expectations can't be violated.

- I expect the problem will be harder than I am anticipating, and will push these systems to their edges
- I expect an agentic system trying out a lot of things on decent compute would make a scratch on the problem
- I don't expect web gui LLMs to solve this
- These are vague, have to add more

Why do you expect this?

What's your reasoning or intuition? This helps you identify which of your assumptions was wrong if the expectation is violated.

- Agentic systems with access to huge compute on the CPU sense is a little underexplored
- I think Web UI won't be able to solve this as they don't have access to that compute
- But the reasoning capabilities are there that a well configured system might make a dent

What would genuinely surprise you?

Describe 1-2 outcomes that would make you stop and say "wait, that's weird." These are the signals you're looking for during the sprint.

- Actually closing the loop on the *entire* $n=12$ (and potentially parts of $n=14$) spectrum and formally proving all the gaps (the UNSAT queries)

FRUITFULNESS PRE-CHECK

Not everything deserves exploring. Before investing time, check whether the best possible outcome of this sprint would actually lead somewhere worth going.

If this surprises you, then what?

Imagine the most interesting possible outcome. What downstream consequences would it have? What research question would it open? If the best case is "huh, neat" and nothing more — reconsider.

The research question it would open would be a deepdive on these different LLM/agent systems and their math abilities. If we are able to solve the problem I am discussing then massive win. It'll give us a highly concrete case study for our AI-for-Science agenda. It would lead to a good research paper both on the AI as scientist POV and just a pure Math POV. The techniques used would need to be genuinely new and novel (from my understanding).

Who would care?

Name a specific community, researcher, or field that would find the result meaningful. If you can't name anyone beyond yourself, the exploration may be intellectually fun but strategically pointless.

- The AI-for-Science community (OpenAI, Google DeepMind, Harmonic, etc.).
- Combinatorial matrix theorists who mapped the $n=11$ and $n=13$ bounds.

Does this connect to our research directions?

Which LossFunk research direction does this fall under? If it doesn't connect to any — is it worth lab time, or is it a personal side interest?

I do think it fits in our [Research direction 2] Creativity in artificial systems. It is more of an application based approach rather than a theoretical undertaking. But I do think it will give us some nice understanding of the paradigm.

TIME BOX

Sprint duration:	1 week
Check-in date:	__30__ / __4__ / ____2026____
Hard stop date:	__8__ / __5__ / ____2026____

At the check-in: is anything surprising emerging? If not, consider stopping early. At the hard stop: make the decision below. No extensions without a new sprint.

What 'done' looks like

What's the minimum you need to run/build/read to know whether something interesting is here? Keep it small. This is a scout mission, not a campaign.

- Setup the agentic workflow with the latest frontier models.
- Give them the context of the n=12 spectrum.
- Will compile the logs of the agent's proposed algorithms and share the results/code by the deadline.
- Some new techniques from what I have seen in literature, some solid cross confirmation from different systems that an approach can work

AFTER THE SPRINT

Fill this in at the end of your time box. Be honest.

What did you actually observe?

Describe what happened. Focus on what matched or didn't match your expectations.

Were your expectations violated?

If yes — how specifically? Which assumption was wrong? If no — that's fine, but it means there's probably not a paper here.

THE DECISION

Pick one. Be disciplined — the whole point of the time box is to force a decision rather than letting exploration drift.

GRADUATE	SHELVE	PIVOT
Surprise found. "I expected X but observed Y. The gap is interesting because:" → Pull out the Research Question Sharpener	No surprise found. Expectations confirmed or results ambiguous. No shame — most explorations end here. File your notes for future reference. What I learned anyway:	Something adjacent is interesting. The original what-if wasn't fruitful, but you noticed something else. Start a new sprint for the new what-if. The new what-if:

The LossFunk Research Flow

Curiosity ("what if...") → **Exploration Sprint** (this template, time-boxed) →
Surprise found?

Yes → **Research Question Sharpener** (one-pager) → **Project**

No → **Shelve** or **Pivot** to new sprint (no guilt, this is how it works)

Lossfunk Research · Exploration is search, not commitment · Most sprints end in 'shelve' — that's
success, not failure